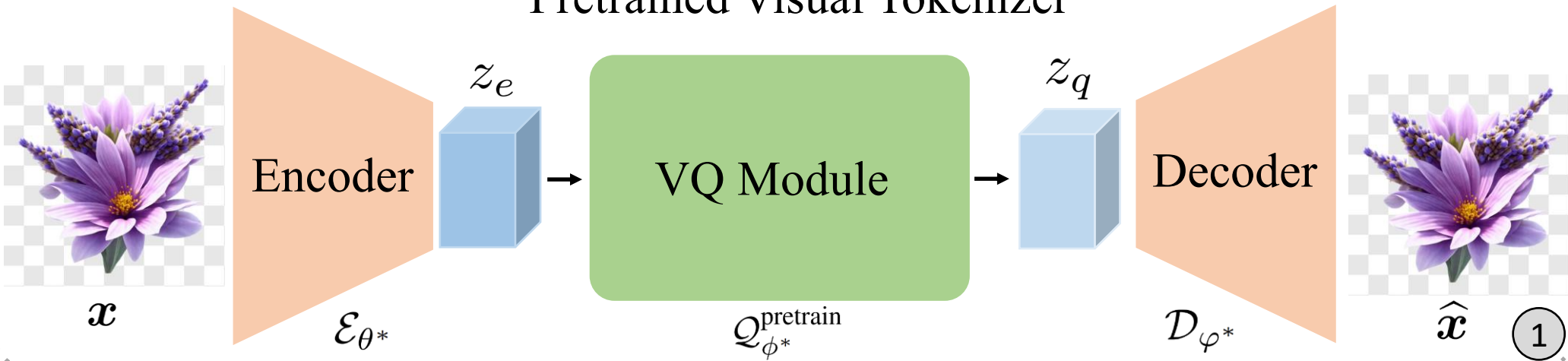


Pretrained Visual Tokenizer



VQ Module Substitution

$$x \rightarrow \mathcal{E}_{\theta^*} \rightarrow z_e \rightarrow \boxed{\mathcal{Q}_{\phi^*}^{\text{new}} \rightarrow z_q(\phi)} \rightarrow \mathcal{D}_{\varphi^*} \rightarrow \hat{x}$$

- Substitute $\mathcal{Q}_{\phi^*}^{\text{pretrain}}$ with $\mathcal{Q}_{\phi}^{\text{new}}$
- Optimize $\mathcal{Q}_{\phi}^{\text{new}}$ to $\mathcal{Q}_{\phi^*}^{\text{new}}$ via $\mathcal{L}_{\text{VQ}}(\phi)$

②

Decoder Adaptation

$$x \rightarrow \mathcal{E}_{\theta^*} \rightarrow z_e \rightarrow \mathcal{Q}_{\phi^*}^{\text{new}} \rightarrow z_q \rightarrow \boxed{\mathcal{D}_{\varphi} \rightarrow \hat{x}(\varphi)}$$

- Initialize $\mathcal{D}_{\varphi}$ by $\mathcal{D}_{\varphi^*}$
- Optimize $\mathcal{D}_{\varphi}$ via $\mathcal{L}_{\text{Decoder}}(\varphi)$

③